Aqueeb Anjum Sunny

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Research Interests

Advanced reactors, Core design, Neutronics, Fuel performance optimization, ML in nuclear engineering

Education

Military Institute of Science and Technology (MIST) July 2025

Bachelor of Science in Nuclear Engineering

CGPA: 3.46 out of 4.00

Research Experience

Undergraduate Thesis January 2024 - May 2025

Neutronics Characteristics Analysis of ALFRED with LFR Reflector Comparison and Evaluation of a Conceptual MPRR Using OpenMC

Supervisor: Prof. Dr. Abdus Sattar Mollah

- Conducted Monte Carlo simulations to analyze the neutronic behavior of the ALFRED reactor core.
- Compared the neutronic performance and activation of lead and LBE as LFR coolants.
- Evaluated reflector materials in an LFR core to assess neutron economy and fuel cycle length.
- Performed preliminary neutronic evaluation of a conceptual Multi-Purpose Research Reactor.

Publications and Conference Projects

1. Shuddho, S. S., **Sunny, A. A.**, & Mollah, A. S. (2026). *Neutronic Performance of Reflector Materials in Lead-Cooled Fast Reactor*. Nuclear Science and Engineering. https://doi.org/10.1080/00295639.2026.2652786 — April 2026
2. Zarin N. Z., **Sunny A. A.**, Shuddho, S. S., and Mollah, A. S. (2024). *Investigation of Neutronics Parameters of Multi-Purpose Research Reactor (MPRR) using VVR-KN Fuel.* International Conference on Electronics and Informatics 2024, AEC, Dhaka — December 2024
3. Dipto, R. R., Shuddho, S. S., **Sunny, A. A.**, & Mollah, A. S. (2024). *Analysis of Neutronics Parameters of Different Annular Fuel Using Monte Carlo Code OpenMC Utilizing JEFF-3.3 and ENDF/B-VIII.0 Nuclear Data Libraries*. Proceedings of the Energy Conference 2023. https://dx.doi.org/10.2139/ssrn.4997514 — October 2024

Technical & Research Skills

Reactor Physics Tools: OpenMC, NJOY, ARMI

Programming Languages: Python (numpy, pandas, matplotlib, scikit-learn), C, MATLAB

Version Control & Environment: Git, Linux, Markdown

Research Skills: Core Design, Monte Carlo Simulations, Data Analysis, Nuclear Data Processing

Portfolio: https://heisen23.github.io/portfolio (Core designs and projects)

Professional Training

Rooppur Nuclear Power Plant, Pabna 7-8 February 2024

- Trained in nuclear power plant operations and safety protocols

Atomic Energy Centre, Dhaka 11-15 February 2024

- Completed Non-Destructive Testing training program

TRIGA Research Reactor, AERE, Savar 5 March 2024

- Studied research reactor operations and neutronics applications

Selected Short Courses & Workshops

Machine Learning Crash Course, Google — September 2025
2025 Nuclear Engineering Summer School, MTV — August 2025
Computational Nuclear Science and Engineering, IAEA — July 2025

Leadership & Outreach

MIST Nuclear Engineering Club, Senior Executive Panel — 2023 - 2024

Organized nuclear engineering events, including a quiz competition for 50+ students, and delivered presentations to 30+ freshmen, introducing them to the field.

Private Tutor (Physics & Mathematics, High School Level) — 2022 - Present

Provided individualized and group tutoring to 20+ higher-secondary students.

Language Proficiency

Bengali: Native
English: Fluent (IELTS Overall Band Score: 8.0)